

ECOSYSTEM VILLAGE PROJECT

Wildroot Village

An ecosystem-management and
environmental learning game

*Grow a home in the forest.
Leave enough wildness for tomorrow.*

BUILD

SIMULATE

LEARN

≈ 3-MINUTE PRESENTATION

CREATED BY

James · age 13

PRESENTED BY

Daniel Bryar



Why I created Wildroot Village

An idea inspired by climate change, global warming and the power of environmental education.

JAMES · CREATOR · AGE 13

PRESENTED BY · DANIEL BRYAR

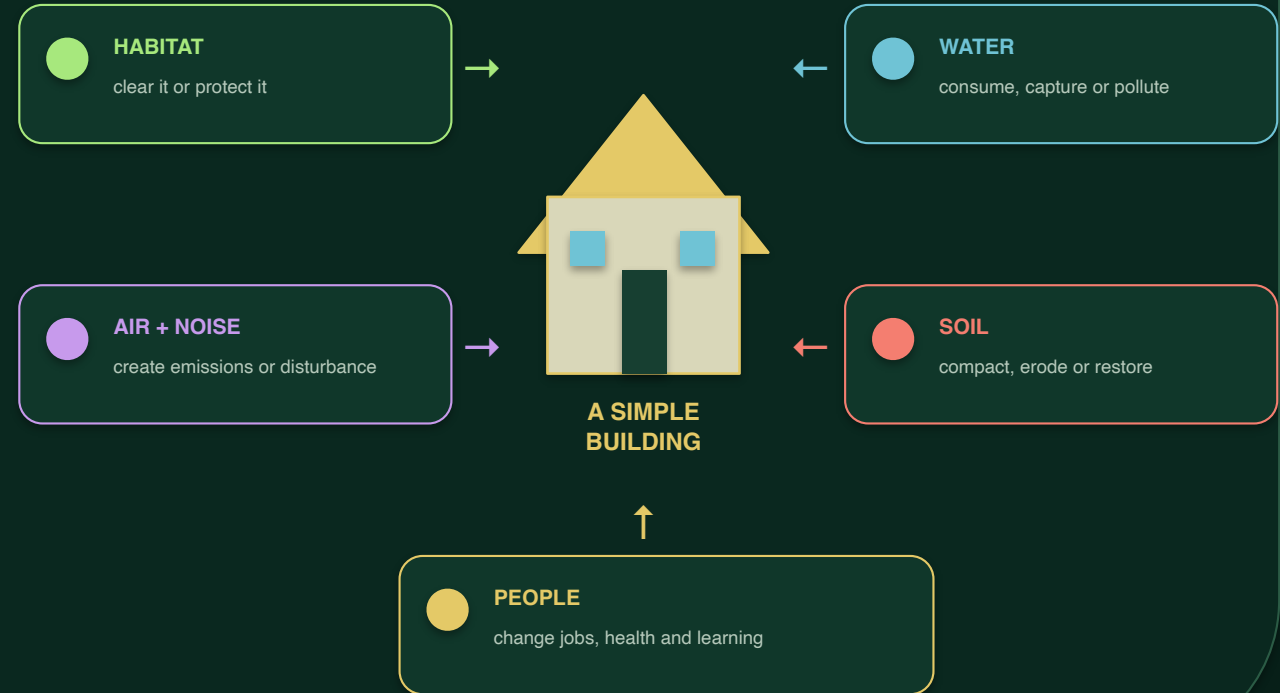
“Yesterday, thinking about climate change and global warming gave me an idea: a game that helps kids see how every building choice can change the environment.”

I want children to learn by making decisions and watching the results—not only by reading facts. Learning is integrated into the game through the Eco Coach, field missions, knowledge checks, inspections and scenarios.

OVERALL GOAL

Create a thriving village with a healthy ecosystem.

ONE SIMPLE BUILDING CAN CHANGE A WHOLE SYSTEM

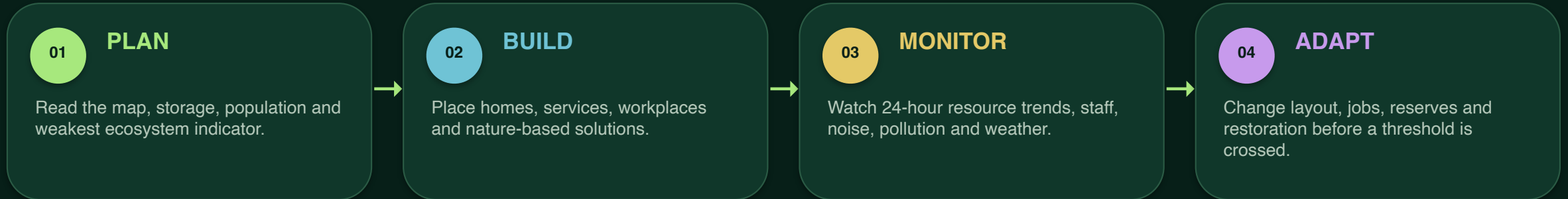


LEARNING IS INTEGRATED INTO PLAY

Eco Coach · field missions · knowledge checks · building inspections · environmental scenarios

The idea: prosper without breaking the system

Every expansion solves one problem while creating new pressure somewhere else.



THE GOAL

Grow a functioning village

Balance population, food, water, timber, stone, health, morale and education—without sacrificing the forest.

THE MAIN TENSION

Short-term output vs. long-term resilience

Industry can speed construction, but resource use, habitat loss, pollution and noise can weaken future production.

THE FAILURE RULE

A weak link can collapse the settlement

The village is lost below 5% overall ecosystem health, or immediately when any one of the six indicators reaches 0%.

100 × 100

forest world

3 minutes

per day at 1×

1× / 2× / 3×

true simulation speed

Freeplay + 8

challenge scenarios

Local

browser save

How the player controls the village

The screen keeps construction, simulation evidence and environmental feedback visible at the same time.

Wildroot Village
Canopy Cross

Day 70 · Summer
14:42 · * Heatwave · 2d 22h left

1 Resources + time

PEOPLE 68 / 72 +1.3/day
FOOD 410/600 -49.1/day
WATER 364/600 -173.6/day
TIMBER 330/600 -0.1/day
STONE 286/600 +3.0/day
ECOSYSTEM 70% -0.2/day

2 Planning desk

3 Living map

4 Ecosystem + Eco Coach

5 Workforce + footprint

PLANNING DESK
Build

All Village Work Nature

Village
Housing for six people, on a modest fuel and waste footprint.
2x2 automatic · 24h · 18 ● 5 Light imp

Storehouse
A 2 x 2 storehouse adding room for 200 food, water, timber and stone items each.
2x2 automatic · 24h · 28 ● 12 Eau

Field Farm
Reliable seasonal food. Two farmers provide normal output; a third raises the total to 2x. Uses water and slowly exhausts unprotected soil.
4x3 3 workers · day shift · 2x full

Village Well
Draws clean groundwater. Too many wells will drain the water table.
1x1 automatic · 24h · 12 ● 12 Draw water

Logging Camp
Loggers rapidly fell trees and break their stumps at the same speed, using a 10x boost inside the 15 x 15 zone. Healthy trees supply 5-10 timber; fire-damaged trees supply 30% less.

WORKERS
46 workers · 23 jobs · 23 available
12 children · School 12/32 · Education 72 · Health 74 · Morale 68 · 56 noise-exposed

LIVING SYSTEM
Ecosystem

The living system is under pressure. Find the weakest indicator and reduce their causes.

Forest cover 76
Wildlife 71
Water quality 67
Soil health 66
Clean air 72
Biodiversity 70

Eco coach 6/20 lessons

WATCH NOW
Water quality · 67/100

Noise-pollution lesson: 56 residents are exposed at home. The full-day average costs 0.38 morale and 0.18 health per day; move loud work at least 6 tiles from housing or remove the source. Water quality is falling 1.05 per day as a full-day average. Limit groundwater demand, capture rain with Rain.

The Green City
The existing village is now in your hands.

Achievement: Growing Pains
Support a population of 25.

Achievement: Full Circle
Survive all four seasons and reach Day 49.

Achievement: Balanced Harvest
Reach Day 31 without the ecosystem ever falling below 70.

Achievement: Bright Future
Provide a school place for every child and raise village education to 70.

Settlement footprint Severe
100 x 100 forest

1 Resources + time

Population, stored materials, /day trends, season, weather and speed controls.

2 Planning desk

Choose Village, Work or Nature buildings; then Inspect or Destroy.

3 Living map

Drag and zoom across the 100 x 100 forest; place only on free clearing tiles.

4 Ecosystem + Eco Coach

Six indicators, overall score, weakest trend, major pressure and best support.

5 Workforce + footprint

Jobs, available workers, children, schooling, health, morale and settlement pressure.

DRAG pan · SCROLL zoom · O/P rotate

SPACE pause · ESC cancel · HOLD tree priority

A living model, not a single score

Six connected indicators reveal weak links that an average could otherwise hide.

LEARN BY WATCHING THE LIVING SYSTEM

Environmental field guide

These principles explain the game's existing cause-and-effect rules. Complete practical missions in any order; learning progress never changes resources or difficulty.

24-hour report - Water quality 67/100

falling 1.05 per day. Clean water is renewed slowly. Wells, farms, people and industry can use or contaminate it faster than rain replaces it.

0/22

CONCEPTS CHECKED

How to read this model

The six indicators and day rates simplify a much more complex reality. Use them to explore feedback and trade-offs; real environmental decisions also require field measurements, local and indigenous knowledge, uncertainty, and impacts beyond the map.

CURRENT EVIDENCE

Six-part systems snapshot

Modelled 24-hour rates

Forest cover

0.35/day

A simplified indicator of wild forest cover, structure and capacity to recover.

Wildlife

0.56/day

A simplified indicator of the abundance and security of wild animal populations.

Water quality

-1.05/day

A simplified indicator combining water cleanliness, availability and watershed health.

Soil health

-0.55/day

A simplified indicator of fertility, soil life, structure and ability to hold water.

Clean air

-0.46/day

A simplified indicator of smoke, dust and the atmosphere's ability to support health.

Biodiversity

+0.58/day

A simplified indicator of variety across species, habitats and ecological relationships.

One connected system

TRY IT

Forest, wildlife, water, soil, air and biodiversity support one another. A high average cannot always rescue one collapsing part.

Real-world connection: Environmental dashboards use several indicators because a single average can hide a polluted river, disappearing species or damaged soil.

Field mission: Reach Day 3 while every ecosystem indicator remains at 70 or higher.

Harvest at regrowth speed

TRY IT

A renewable resource is sustainable only when it can regrow at least as quickly as it is harvested.

Real-world connection: Foresters compare harvest with growth over many years and protect old, dead and habitat trees that timber totals do not describe.

Field mission: Use a Forester, or connect a Logging Camp to a nearby Wood Farm.

LIVE 24-HOUR FIELD GUIDE

TRY IT

Work with the water cycle

TRY IT

The Green City

The existing village is now in your hands.

THE SIX INDICATORS

Forest cover

76

Habitat, carbon and recovery capacity

Wildlife

71

Abundance and security of animal populations

Water quality

67

Cleanliness, supply and watershed health

Soil health

65

Fertility, soil life and structural stability

Clean air

72

Smoke, dust and atmospheric capacity

Biodiversity

70

Species, habitats and ecological relationships

CAUSE AND EFFECT

BUILDINGS + PEOPLE

use resources; create output, noise and pollution

→

WEATHER + SEASONS

change demand, production, fires, floods and growth

→

24-HOUR RATES

combine daytime staffing with overnight processes

→

FEEDBACK

changes food, health, morale and long-term resilience

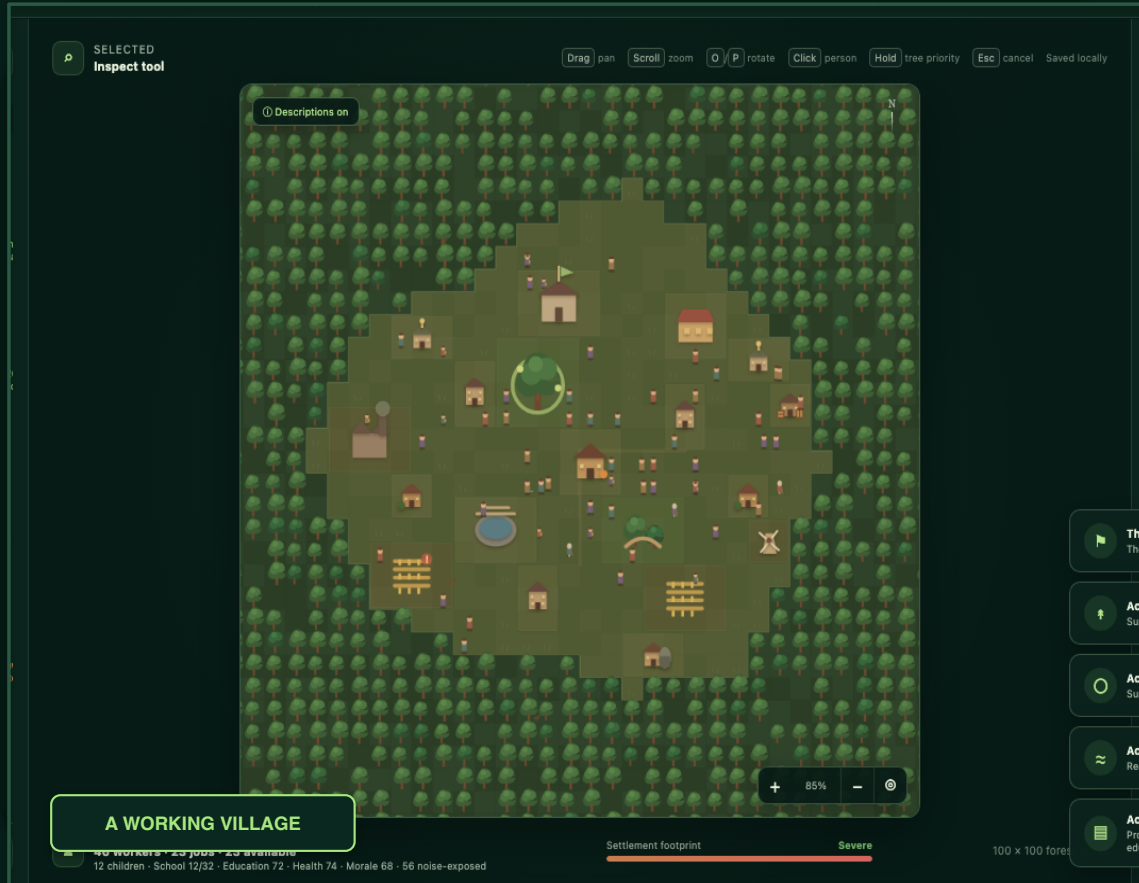
COLLAPSE: <5% OR ANY 0%

WILDROOT VILLAGE · ECOSYSTEM VILLAGE PROJECT

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People, work and the resource economy

Residents have homes, ages, jobs and journeys; buildings operate on different schedules and capacities.



PEOPLE + LIFE

Freeplay starts with 10 adult founders. The population includes children, adults and elders. Children become adults after 7 in-game days; residents have persistent homes and a hidden natural lifespan of 40–60 days.

WORK + MOVEMENT

Staffed buildings operate 07:00–18:30. Residents walk between home, work, parks, school and storage; workers carry 1–3 items per supply trip. Automatic buildings continue overnight.

STORAGE + SUPPLY

The Founders' Hearth stores 400 of each material. Every Storehouse adds 200 food, water, timber and stone capacity. Production above the cap is discarded, so overnight reserves matter.

THREE IMPORTANT TRADE-OFFS

FARMS

1 farmer = 0.5x; 2 = 1x; 3 = 2x output—and 2x water use and pressure.

FORESTRY

Logging is fast nearby; Wood Farms regrow 16 plots in 5 days. Full storage pauses cutting.

SITING

Industry near crops reduces yield; loud buildings near occupied homes lower morale and health.

Eight scenarios turn systems into problems

Each village begins with a different crisis, four measurable goals and a named environmental learning focus.

8 STARTING VILLAGES

Wildroot Village

Grow a home in the forest.
Leave enough wildness for tomorrow.

New clearing

Eight scenarios

Take charge of an existing village. Eight settlements teach different ecological problems; three begin below 40% with an active restoration emergency.

Winter Watch

DAY 14
18 PEOPLE
81% ECOSYSTEM
BALANCED

The Dry River

DAY 19
22 PEOPLE
88% ECOSYSTEM
HARD

Timber Debt

DAY 27
28 PEOPLE
85% ECOSYSTEM
BALANCED

Smoke Valley

DAY 42
42 PEOPLE
55% ECOSYSTEM
HARD

After the Fire

DAY 15
23 PEOPLE
34% ECOSYSTEM
BALANCED

The Poisoned River

DAY 50
34 PEOPLE
32% ECOSYSTEM
HARD

The Silent Fields

DAY 54
26 PEOPLE
39% ECOSYSTEM
HARD

The Green City

DAY 70
88 PEOPLE
70% ECOSYSTEM
HARD

SCENARIO - BALANCED DIFFICULTY

After the Fire

A severe fire stripped the upper catchment. Rain now carries ash and soil through a village whose ecosystem is below 40% and still falling by about 1% each day. Most fire-damaged trees remain permanently black and leafless and contain 30% less usable timber.

Existing village

Emberford - 9 buildings

Starting point

Day 13 - Summer - 22 residents

Living system

34% ecosystem - Steady rain

Children

5 - Grow into adults in 7 days

Collapse rule

Lose below 5% overall or when any indicator reaches 0%

Active decline

-1.0 ecosystem/day - Post-fire erosion and habitat loss

Stop the crisis

Retire every Logging Camp, maintain two Rain Gardens and build a Compost Yard.

Learning focus

Post-fire recovery: protect damaged standing trees where possible, stop further disturbance, slow runoff and restore soil, vegetation and habitat together rather than replacing only the lost timber.

SCENARIO GOALS

1 Retire every Logging Camp

2 Maintain two Rain Gardens and a Compost Yard

3 Restore forest cover and soil health to 65+

4 Reach Day 30 with a 60+ ecosystem

Share results

Thyphar and Emberford

RESTORATION BRIEFING

FIVE CHALLENGE VILLAGES

Winter Watch · The Dry River · Timber Debt · Smoke Valley · The Green City

They test seasonal reserves, drought, renewable forestry, clean industry and dense sustainable growth.

THREE ACTIVE EMERGENCIES

After the Fire · The Poisoned River · The Silent Fields

Each starts below 40% and loses 1.0 ecosystem point/day until its containment actions are completed.

DYNAMIC RISK

Weather ranges from ~2-hour thunderstorms and ~12-hour rain to 3-day heatwaves, 4-day cloud systems and 7-day droughts. Random events arrive every ⅓–7 days.

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Learning is built into the decisions

Players learn by observing consequences, testing ideas and explaining what the model leaves out.

THE LEARNING LAYER

- 20 field missions** — demonstrate principles through normal gameplay; every lesson links to a real-world example.
- 22 knowledge checks** — pause time, explain correct and incorrect answers, and give no resource reward or penalty.
- Learning everywhere** — Eco Coach advice, building inspections, events and scenario briefings all include a learning lens.
- Model literacy** — the game names its simplifications and points to field evidence, local and Indigenous knowledge and uncertainty.
- Saved progression** — learning and achievements persist locally with the current village.

FINAL CHAPTER

Day 100 · nature buildings · every indicator 90+ · 100% ecosystem

BUILD →
MEASURE →
ADAPT → LEAVE
WILDNESS FOR
TOMORROW

